

# ILINA Junior Research Fellowship 2026

## Runtime Safety Governors: Activation Steering as a Control API for Deployed Language Models

| Track         | Seminar Period | Project Period           |  |
|---------------|----------------|--------------------------|--|
| Technical JRF | Apr–Jun 2026   | Aug–Nov 2026<br>(20h/wk) |  |

### 1. Background & Motivation

The dominant paradigm in AI safety is training-time alignment: techniques such as Reinforcement Learning from Human Feedback (RLHF) and Constitutional AI attempt to bake safe behaviour into model weights before deployment. While effective in many settings, this paradigm is fundamentally static. Model weights are frozen once deployed; the world, adversaries, and deployment contexts are not.

This creates what we term the Static Alignment Trap: an over-reliance on fixed weight configurations to ensure safe behaviour across arbitrarily diverse and adversarial runtime conditions. Three concrete failure modes illustrate the gap:

- **Sleeper Agents** (Hubinger et al., 2024): Models that behave safely during training and evaluation but activate deceptive behaviour upon encountering specific deployment-time triggers. No amount of pre-deployment testing reliably surfaces these.
- **Translation Jailbreaks**: Safety alignment trained predominantly on English data demonstrably fails when adversaries submit harmful requests in low-resource languages. A model may refuse a request in English and comply with the identical request in Swahili.
- **Mid-Generation Deception**: In agentic and long-context settings, models can enter deceptive reasoning chains during a single forward pass. The retraining cycle (detect → log → fine-tune → redeploy) operates on timescales of weeks; the harm can occur in milliseconds.

My prior work (Ashioya, 2026) provides mechanistic grounding for this concern. In a mechanistic interpretability study on GPT-2 Small, we achieved 88% probe accuracy (ROC-AUC 0.949) in detecting unfaithful Chain-of-Thought from residual stream activations, identified 23 causally implicated circuit components, and demonstrated ~72% suppression of deceptive reasoning via steering vectors. Critically, Layer 0 MLP emerged as the dominant causal component, while attention heads L1H10 and L6H6 were identified as shortcut and harmful heads respectively.

This project extends that foundation: rather than merely detecting unfaithful reasoning post-hoc, we ask whether activation steering can serve as a reliable runtime Control API — a steering wheel, not just a diagnostic.

## 2. Problem Statement

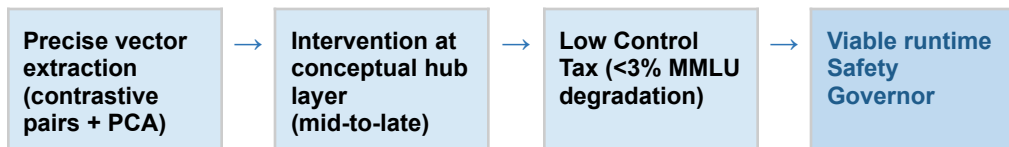
*Core Problem: Current AI safety operates as a one-time vaccination rather than an immune system. We need runtime control mechanisms that can intercept, detect, and correct unsafe model behaviour during inference — without retraining.*

More specifically, the field lacks:

- A principled framework for treating activation steering as a deployable Control API rather than a research curiosity
- Quantitative understanding of the Control Tax: the capability-safety tradeoff when steering vectors are clamped at inference time
- Evidence on whether safety steering vectors generalise across languages, particularly to low-resource African languages such as Swahili
- Mechanistic grounding for where in the network (which layers, which components) runtime interventions are most effective and least destructive

## 3. Theory of Change

The project operates on a simple causal logic: if we can precisely identify the latent directions corresponding to unsafe reasoning modes, and if those directions are sufficiently disentangled from directions encoding general capability, then inference-time intervention becomes feasible without unacceptable utility loss.



The cross-lingual generalisation experiment is the critical stress test. If English-extracted safety vectors fail to steer Swahili prompts, this is evidence that current safety alignment is language-specific rather than concept-level — a publishable null result with direct policy implications for AI deployment in multilingual contexts, particularly across Africa.

Conversely, if vectors do transfer, this opens a pathway to low-cost safety coverage for low-resource languages without requiring expensive multilingual safety datasets.

## 4. Research Questions

### RQ1

Can we extract activation vectors that reliably represent deceptive reasoning and instruction non-compliance modes in open-weight LLMs, and do these vectors maintain directional stability across diverse prompt distributions?

|               |                                                                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ2           | What is the Control Tax of inference-time safety steering? Specifically, how does MMLU accuracy, response fluency (perplexity), and targeted behaviour change as a function of steering coefficient magnitude?          |
| RQ3           | Do safety steering vectors extracted from English data generalise to Swahili prompts? If so, at which layers? If not, what does the failure mode reveal about the architecture of cross-lingual safety representations? |
| RQ4 (Stretch) | Can a lightweight runtime classifier, operating on residual stream activations, dynamically trigger steering interventions only when unsafe reasoning is detected, minimising the Control Tax?                          |

## 5. Methodology

### 5.1 Models & Infrastructure

Primary experimental model: Llama-3-8B (instruction-tuned). Chosen for scale (realistic deployment size), open weights, and existing interpretability literature. GPT-2 Small used for rapid prototyping, consistent with prior CoT faithfulness work.

Tooling: TransformerLens (activation extraction and patching), nnsight (alternative patching), Colab GPUs, Weights & Biases (experiment tracking).

### 5.2 Stage 1: Vector Extraction

We will construct contrastive pair datasets for three target behaviours:

- **Deceptive Reasoning**: pairs of prompts where the model reasons faithfully vs. unfaithfully (leveraging the dataset from our CoT faithfulness work)
- **Instruction Non-Compliance**: pairs where the model follows vs. evades instructions
- **Harmful Compliance**: pairs from JailbreakBench (Chao et al., 2024) — harmful vs. refused

Vector extraction methods (in order of expected precision):

1. Difference-in-Means (Rimsky et al., 2024): average activation difference between contrastive pairs at each layer
2. PCA on contrastive activations: captures dominant direction of variation between safe/unsafe classes
3. Linear probe weights: train a linear classifier, use decision boundary normal as the steering direction

We require at least 80–100 contrastive pairs per behaviour to avoid high-variance vectors (Weij et al., 2024). Dataset will be bootstrapped using existing benchmarks and synthetic generation.

### 5.3 Stage 2: Intervention & Control Tax Measurement

For each extracted vector  $v$ , we apply activation addition at layer  $L$  during inference:

$$A'_L = A_L + \alpha \times v$$

We sweep over: layers (every 4th layer), coefficients  $\alpha \in \{1, 2, 5, 10, 20\}$ , and token positions (last prompt token vs. all tokens).

Control Tax is measured along three axes:

- **Capability:** MMLU 5-shot accuracy (delta from unsteered baseline)
- **Fluency:** Perplexity on WikiText-103 (measures coherence degradation)
- **Targeted suppression:** Attack Success Rate (ASR) on JailbreakBench harmful queries; compliance rate on instruction non-compliance dataset

The acceptable threshold is defined as: targeted suppression >70% with MMLU delta <3%. This operationalises the concept of a “surgical” vs. “lobotomising” intervention.

### 5.4 Stage 3: Cross-Lingual Generalisation

This is the most novel and Africa-relevant component of the project.

Step 1 — Identify the conceptual hub layer via Representation Similarity Analysis: run parallel forward passes on semantically identical English/Swahili prompt pairs, extract residual stream activations at every layer, compute cosine similarity between language pairs. The layer at which similarity peaks is the conceptual hub.

Step 2 — Inject the English-extracted vector at the identified hub layer during a Swahili forward pass. Measure transfer efficacy using the same three Control Tax axes.

Hypothesis: Vectors extracted from the hub layer will show positive transfer because this is where language-agnostic concept representations are formed (Turner et al., 2023; Scalena et al., 2024). Vectors extracted from early layers will not transfer, as those layers are token-bound to English syntax.

Swahili test data: we will construct 100 Swahili harmful/benign prompt pairs, leveraging existing Swahili NLP infrastructure from the Sauti-Ya-Kenya project (MsingiAI). This is a direct synergy between active projects.

## 6. Timeline

| Phase | Period | Key Deliverables |
|-------|--------|------------------|
|-------|--------|------------------|

|                 |                     |                                                                                                                                                                                                                                   |
|-----------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-Work</b> | Apr–Jul 2026        | <ul style="list-style-type: none"> <li>• Contrastive pair dataset construction (300+ pairs)</li> <li>• Replicate Rinsky et al. (2024) on Llama-3-8B</li> </ul>                                                                    |
| <b>Stage 1</b>  | Aug Wk 1–2          | <ul style="list-style-type: none"> <li>• Vector extraction (DIM + PCA + probe)</li> <li>• Layer-wise stability analysis</li> <li>• GitHub repo setup with reproducible experiments</li> </ul>                                     |
| <b>Stage 2</b>  | Aug Wk 3 – Sep Wk 2 | <ul style="list-style-type: none"> <li>• Intervention sweep (<math>\alpha</math>, layer, token position)</li> <li>• Control Tax measurement (MMLU, perplexity, ASR)</li> <li>• Identify optimal steering configuration</li> </ul> |
| <b>Stage 3</b>  | Sep Wk 3 – Oct Wk 3 | <ul style="list-style-type: none"> <li>• Cosine similarity layer mapping (EN/SW)</li> <li>• Cross-lingual injection experiments</li> <li>• Transfer efficacy analysis</li> </ul>                                                  |
| <b>Writeup</b>  | Oct Wk 4 – Nov 2026 | <ul style="list-style-type: none"> <li>• Technical report + paper draft</li> <li>• Results blog post for Unsupervised Insights</li> <li>• ICLR / NeurIPS submission prep (stretch)</li> </ul>                                     |

## 7. Expected Outcomes & Contributions

### Primary Contributions

- **Runtime Safety Governor Protocol:** A documented, reproducible methodology for deploying activation steering as a lightweight safety intervention at inference time, with clear Control Tax characterisation.
- **Cross-Lingual Safety Transfer Findings:** Either positive transfer evidence (enabling low-cost Swahili safety coverage) or a null result demonstrating that safety alignment is English-specific both are publishable and policy-relevant.
- **Open Dataset:** A curated English/Swahili contrastive safety pair dataset, the first of its kind for East African languages in an interpretability context.

### Secondary Contributions

- Layer-wise Control Tax characterisation for Llama-3-8B (adds to sparse existing literature on this model family)
- Integration with CoT faithfulness prior work: if L0MLP dominance generalises from GPT-2 to Llama, this is a significant cross-model finding
- Policy brief for African AI governance stakeholders on multilingual safety coverage gaps

## 8. Related Work

## 8.1 Activation Steering Foundations

Turner et al. (2023) introduced Activation Addition (ActAdd), demonstrating inference-time control over high-level properties (sentiment, topic) via contrastive steering vectors added to the residual stream. Rimsky et al. (2024) scaled this to Llama-2-13B via Contrastive Activation Addition (CAA), achieving strong results on corrigibility and sycophancy suppression. Zou et al. (2023) formalised Representation Engineering as a top-down approach to AI transparency, identifying linear directions for truthfulness, emotion, and power-seeking.

## 8.2 The Control Tax Problem

The alignment-capability tradeoff under steering (colloquially the Control Tax) is documented in Weij et al. (2024) and Soo et al. (2025). Key finding: steering at early layers and with high coefficients degrades fluency disproportionately; late-layer interventions provide better tradeoffs. Suri et al. (2025) further show that dynamic, late-layer interventions outperform static patches. The Rogue Scalpel paper (arXiv:2603.24543) provides a sobering counterpoint: random steering directions can break refusal mechanisms, suggesting that imprecision in vector extraction carries real safety risk.

## 8.3 Cross-Lingual Safety & Multilingual Representations

Safety alignment trained on English demonstrably fails under cross-lingual transfer (Bhardwaj et al., 2025; Zhao et al., 2025). Tang et al. (2024) identified language-specific neurons encoding language identity, providing a mechanistic basis for understanding why safety fails cross-lingually. Scalena et al. (2024) showed that language adaptation via steering vectors between English and Italian latent subspaces achieves performance at or above fully fine-tuned models. Wang et al. (2025) proposed cross-lingual activation steering to improve multilingual reasoning, finding that reinforcing processing in the shared semantic space improves smaller models especially. This directly motivates our Swahili transfer experiments.

## 8.4 Positioning Relative to Prior Work

This project is novel along two axes: (1) the explicit framing of activation steering as a runtime Control API with quantified Control Tax, rather than an exploratory interpretability tool; and (2) the first systematic cross-lingual safety transfer study targeting a low-resource African language (Swahili), grounded in existing African NLP infrastructure.

## 9. Key References

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